LECTURE NOTES

Business Finance

Chapter 11. Risk and Return

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Chapter Roadmap

11

§	Topic	What you should be able to do
11.1	Expected Returns and Variances	Compute an expected return and its variance from a set of states
11.2	Portfolios	Compute a portfolio's expected return, and see why its risk is lower
11.3	Announcements, Surprises, and Expected Returns	Split an actual return into the part expected and the part that surprised
11.4	Risk: Systematic and Unsystematic	Tell the two kinds of risk apart by what they affect
11.5	Diversification and Portfolio Risk	Say which risk diversification removes and which it cannot
11.6	Systematic Risk and Beta	Read a beta, and compute the beta of a portfolio
11.7	The Security Market Line	Derive the CAPM from one no-arbitrage argument, and apply it
11.8	The SML and the Cost of Capital	Use the required return as the discount rate for a project

The chapter answers the question Chapter 10 left open: **not how much risk an asset has, but how much of its risk anyone has to be paid for.**

11.1

Expected Returns and Variances

- Looking forward instead of backward
- Probability-weighted returns
- Variance measured against the expectation

Expected Return

11.1

A probability-weighted average over the states that could occur

Chapter 10 averaged what did happen. **This chapter averages what might, weighting each outcome by its chance.**

$$E(R) = p_1R_1 + p_2R_2 + \ldots + p_nR_n$$

State	Chance	Return on A	Return on B
Recession	0.50	−20%	+30%
Boom	0.50	+70%	+10%
Expected return		**25%**	**20%**

$E(R_a)$ = **0.50 × (−20%) + 0.50 × 70% = 25%**

$E(R_\beta)$ = **0.50 × 30% + 0.50 × 10% = 20%**

Stock A has the higher expected return. It is not therefore the better buy — that depends on its risk.

Variance of an Expected Return

11.1

How far the states sit from the expectation

$$\sigma^2 = p_1(R_1 - E(R))^2 + \ldots + p_n(R_n - E(R))^2$$

Stock A — E(R) = 25%

State	Return	p × (R − 25%)²
Recession	−20%	0.50 × (−0.45)²
Boom	+70%	0.50 × (0.45)²
Variance		**0.2025**

Stock B — E(R) = 20%

State	Return	p × (R − 20%)²
Recession	+30%	0.50 × (0.10)²
Boom	+10%	0.50 × (−0.10)²
Variance		**0.0100**

	Expected return	Variance	Standard deviation
Stock A	25%	0.2025	45%
Stock B	20%	0.0100	10%

11.2

Portfolios

- Expected return adds up as a weighted average
- Risk does not
- Where the difference comes from

A Portfolio's Expected Return

11.2

Half in A, half in B — two routes to the same answer

$$E(R_P) = w_1E(R_1) + w_2E(R_2) + \ldots$$

Route 1 — average the two expected returns

$E(R_p) = 0.50 \times 25\% + 0.50 \times 20\%$

$= 12.5\% + 10\% = 22.5\%$

Route 2 — work out the portfolio state by state

State	Chance	Portfolio
Recession	0.50	+5%
Boom	0.50	+40%
Expected		**22.5%**

Recession: $0.50 \times (-20\%) + 0.50 \times 30\% = +5\%$

Boom: $0.50 \times 70\% + 0.50 \times 10\% = +40\%$

Expected: $0.50 \times 5\% + 0.50 \times 40\% = 22.5\%$ (= route 1)

Both routes give 22.5%, and they always will.

Expected returns are weighted averages, so it makes no difference whether you average first and combine, or combine first and average.

Portfolio Risk Is Not a Weighted Average

11.2

Stock A's standard deviation is 45% and stock B's is 10%. **A half-and-half portfolio does not have a standard deviation of 27.5%.**

Compute it properly, from the portfolio's own states

State	Chance	Portfolio return	$p \times (R - 22.5\%)^2$
Recession	0.50	+5%	$0.50 \times (-0.175)^2$
Boom	0.50	+40%	$0.50 \times (0.175)^2$
Variance			**0.030625**

	If risk were a weighted average	What it actually is
Standard deviation	**$0.50 \times 45\% + 0.50 \times 10\% = 27.5\%$**	**$\sqrt{0.030625} = 17.5\%$**

Practice Question

11.1

Expected return and standard deviation

Two stocks, three states of the economy, and the probabilities do not come out even. **Work in decimals and keep four places until the last step.**

State of the economy	Probability	Return on C	Return on D
Boom	0.25	+40%	+15%
Normal	0.55	+12%	+10%
Recession	**? = 0.20**	**−20%**	**−5%**

Answer all four parts

(a) What is the probability of a recession? Say how you know.

→ **1 − 0.25 − 0.55 = 0.20 (probabilities sum to 1)**

(b) Compute the expected return on each stock.

→ **E(R): C = 12.60%, D = 8.25%**

(c) Compute the variance and the standard deviation of each stock's return.

→ **Var: C = 0.040044, D = 0.004819; SD: C = 20.01%, D = 6.94%**

Practice Question — Solution

(a) Probabilities must sum to 1, so P(recession) = 1 − 0.25 − 0.55 = **0.20**

(b) Expected returns

	Stock C	Stock D
0.25 × boom	0.1000	0.0375
0.55 × normal	0.0660	0.0550
0.20 × recession	−0.0400	−0.0100
E(R)	**12.60%**	**8.25%**

(c) Variance and standard deviation

$p \times (R - E(R))^2$	Stock C	Stock D
Boom	0.018769	0.001139
Normal	0.000020	0.000168
Recession	0.021255	0.003511
Variance	0.040044	0.004819
Standard deviation	**20.01%**	**6.94%**

TI BA II Plus

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2nd DATA   2nd CLR WORK
40  ENTER ↓  25 ENTER ↓
12  ENTER ↓  55 ENTER ↓
20 +/- ENTER ↓  20 ENTER ↓
2nd STAT   ↓ ↓
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$\bar{x} = 12.60$ $\sigma x = 20.01$

11.3

Announcements, Surprises, and Expected Returns

- Why prices move when news arrives
- The part already in the price, and the part that is not
- Only the surprise matters

Expected and Unexpected Returns

11.3

Every actual return has two pieces

$$R = E(R) + U$$

actual return = the part the market already expected + the surprise

E(R)

What the market was already pricing in before the year started.
It is built into the price you paid, so it earns you nothing extra when it arrives.

U

Everything that was not anticipated.
It averages to zero over many periods — by definition, since anything with a predictable sign would already be in E(R).

Worked: the market expects stock A to return 25% this year, and it actually returns 30%. Then E(R) = 25% and U = +5%.
The investor is five points better off than expected — and that five points is the only part nobody could have planned for.

11.4

Risk: Systematic and Unsystematic

- Two kinds of surprise
- One hits everything, one hits a single firm
- The distinction that decides what gets paid for

Two Kinds of Risk

11.4

The difference is how many firms the surprise reaches

	Systematic risk	Unsystematic risk
Also called	Market risk, nondiversifiable risk	Unique risk, asset-specific risk
Who it affects	Nearly every asset, to some degree	One firm, or at most a small group
Examples	Interest rates, inflation, GDP growth, a change in tax law	A strike at one plant, a failed product, a lawsuit, a fire
In a large portfolio	Remains — every holding is exposed to it	Cancels out — one firm's bad luck offsets another's good

11.5

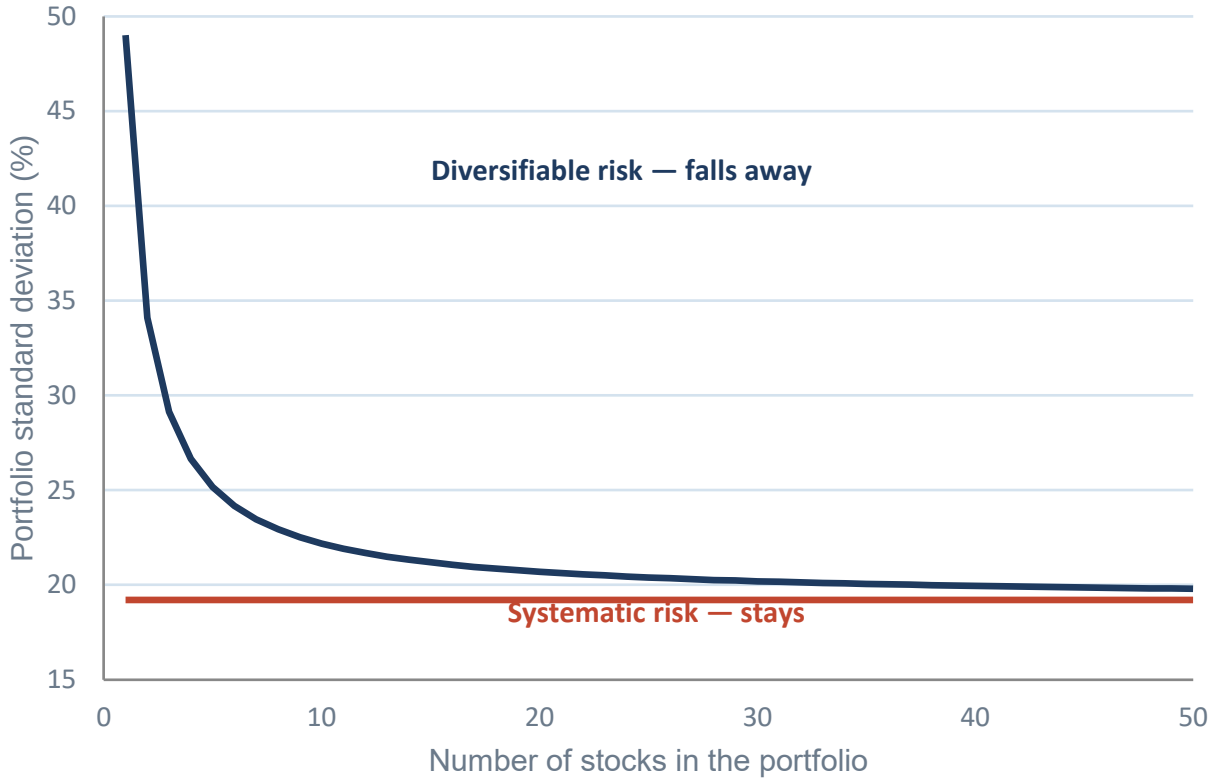
Diversification and Portfolio Risk

- What happens as a portfolio grows
- The part that falls away
- The floor it cannot fall below

Diversification

11.5

Portfolio risk as holdings are added, one stock at a time



Stocks held	Standard deviation
1	49.0%
5	25.2%
10	22.2%
30	20.2%
50	19.8%

Most of the benefit arrives early. Going from one holding to ten removes more than half the risk. Going from thirty to fifty removes almost none.

Schematic. The gap between the two lines is the risk that averaging across holdings removes.

What Diversification Can and Cannot Do

11.5

The conclusion that the whole chapter turns on

It removes unsystematic risk	Spread holdings across enough firms and one firm's bad luck is offset by another's good. The cost of doing this is close to zero.
It cannot remove systematic risk	Every holding is exposed to interest rates, inflation and the business cycle. Averaging across them leaves that exposure untouched.
So only systematic risk is rewarded	A market will not pay anyone for bearing a risk they could have shed for free. The reward attaches to the part that cannot be shed.

11.6

Systematic Risk and Beta

- Measuring the risk that cannot be diversified away
- What a beta of 1 means
- The beta of a portfolio

Beta

11.6

How much of the market's movement an asset picks up

Standard deviation measures all of an asset's risk.
Beta measures only the part that moves with the market — the part that survives diversification.

Reading a beta

Beta	Means
$\beta = 1$	Moves with the market, one for one
$\beta > 1$	Amplifies the market's moves
$0 < \beta < 1$	Dampens them
$\beta = 0$	Unaffected by the market

Roughly where real firms sit

Kind of business	Typical beta
Utilities, staple foods	**below 1 (≈ 0.5)**
Broad market index	**1.0**
Airlines, luxury goods, technology	**above 1 (≈ 1.5)**

Why those businesses. People buy electricity and bread in a recession, so a utility's earnings barely track the cycle. People postpone flights, handbags and new systems, so those earnings swing more than the economy does — and so do their share prices.

A high standard deviation does not imply a high beta. A gold miner can be wildly volatile and barely track the market at all.

The Beta of a Portfolio

11.6

Here the weighted average does work

$$\beta_P = w_1\beta_1 + w_2\beta_2 + \ldots$$

Holding	Weight	Beta	Contribution
Stock A	40%	1.60	0.40 × 1.60 = 0.64
Stock B	60%	0.80	0.60 × 0.80 = 0.48
Portfolio	**100%**	**1.12**	**0.64 + 0.48 = 1.12**

A portfolio's beta is a weighted average of its holdings' betas. A portfolio's standard deviation is not a weighted average of theirs.

11.7

The Security Market Line

- One argument that prices every asset
- Reward-to-risk must be the same everywhere
- The capital asset pricing model

Reward-to-Risk Must Be the Same for Everything

11.7

The argument the whole model rests on

$$Reward - to - risk\ ratio = \frac{E(R_i) - R_f}{\beta_i}$$

extra expected return, per unit of systematic risk, with the risk-free rate at 8%

	Expected return	Beta	Reward-to-risk
Asset A	22%	1.60	**(22% – 8%) / 1.60 = 8.75%**
Asset B	13%	1.20	**(13% – 8%) / 1.20 = 4.17%**

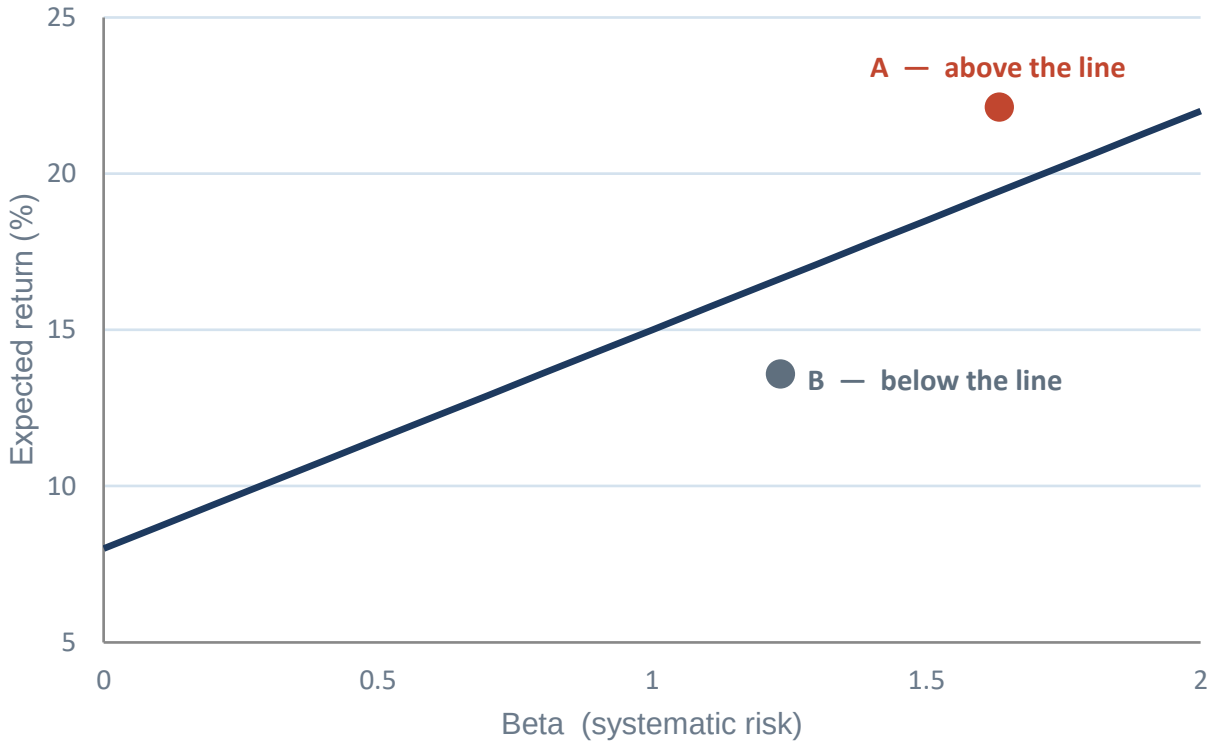
This cannot last. A is paying 8.75% for every unit of systematic risk and B is paying 4.17% for the same thing. Investors sell B and buy A. A's price rises and its expected return falls; B's does the opposite — until the two ratios meet.

So in equilibrium every asset has the same reward-to-risk ratio. **Write that condition down and the capital asset pricing model falls out of it.**

The Security Market Line

11.7

Every correctly priced asset lies on one straight line



Risk-free rate at β = 0, the market portfolio at β = 1.

	Offers	SML says	Verdict
Asset A	22%	19.2%	cheap
Asset B	13%	16.4%	dear

Above the line is undervalued. The asset offers more than its beta requires, so buyers compete for it and the price rises until it sits on the line.

The slope is the market risk premium. Here it is 7%: each extra unit of beta earns seven more points of expected return.

The Capital Asset Pricing Model

11.7

The security market line, written as an equation

$$E(R_i) = R_f + \beta_i \times [E(R_M) - R_f]$$

Term	Is	In the example
Rf	The return with no systematic risk	8%
$E(R_m) - Rf$	The market risk premium — the slope	7%
β_i	How much of that risk this asset carries	varies by asset

Required return at each beta

Beta	8% + β × 7%
0.0	**8% + 0.0 × 7% = 8.0%**
1.0	**8% + 1.0 × 7% = 15.0%**
1.6	**8% + 1.6 × 7% = 19.2%**

Worked: a portfolio

40% in a stock with β = 1.60, 60% in one with β = 0.80.

β_p = **0.40 × 1.60 + 0.60 × 0.80 = 1.12**

$E(R_p)$ = 8% + 1.12 × 7% = 15.84%

11.8

The SML and the Cost of Capital

- From a required return to a discount rate
- Why the project's risk is what matters
- The link back to Chapters 8 and 9

The Required Return Is the Discount Rate

11.8

Closing the loop that Chapter 8 opened

Chapters 8 and 9 discounted project cash flows at a required return and never said where it came from. **The SML is where it comes from.**

The rule	A project is worth undertaking if its expected return exceeds what the market requires for that level of systematic risk — that is, if it plots above the SML.
Use the project's risk, not the firm's	A stable utility opening a software venture must discount that venture at a software beta. Using the firm's own low beta would make almost any project look acceptable.
Where the number goes	The required return from the SML becomes the discount rate in the NPV calculation. Positive NPV and plotting above the SML are the same test.